

研究方向

Graph Learning

Graph Neural Network; Geometric Deep Learning; Graph-Related LLMs

AI for Science

Graph for Science, Drug Discovery, Molecular Representation Learning, Computational Electromagnetics

Industry Application

AI for IT Operations; Fake News Detection; Fraud Analysis and Detection;

教育背景

- 2018–2021 计算机科学 博士.
佛罗里达州立大学, 美国
- 博士毕业论文: 《Mining on Graphs: Graph Neural Network and Applications》
 - 博士导师: Jiawei Zhang(张嘉洧), Ph.D.
- 2015–2017 计算机科学 硕士.
伍斯特理工学院, 美国
- 硕士导师: Xiangnan Kong(孔祥南), Ph.D.
- 2011–2015 软件工程学士, 法学双学士.
南京大学, 中国

工作经历

- 2025–至今 助理教授 (**Tenure-track**), 南京大学-智能科学与技术学院, 苏州, 中国.
- 特聘研究员、博士生导师
- 2022–2025 高级研究员, 华为中央研究院-先进计算与存储实验室, 上海, 中国.
- 天才少年计划
- 2018–2021 研究助理, 佛罗里达州立大学-计算机科学系, 佛罗里达, 美国.
- 导师: Jiawei Zhang, Ph.D.
- 2020, 2021 科研实习生, NEC 美国研究院-深度学习组, 普林斯顿, 美国.
- 导师: Lu-An Tang, Ph.D. and Wei Cheng, Ph.D.
- 2019 科研实习生, 京东硅谷研究院-机器学习组, 加利福尼亚, 美国.
- 导师: Liefeng Bo, Ph.D. and Bo Liu, Ph.D.
- 2018 科研实习生, 京东数科-风险管理部, 北京, 中国.
- 导师: Hao Zhu, Ph.D.
- 2016–2017 软件开发实习生, Dealertrack Technologies, 康涅狄格 美国.
- 2014–2015 软件开发实习生, IBM CDL 中国研发实验室, 上海, 中国.

奖项与荣誉

2025	公有云业务部总裁奖	华为技术有限公司
2025	终端云服务总裁奖	华为技术有限公司
2025	华为 2012 实验室总裁奖	华为技术有限公司
2024	新型电力系统智能仿真器驭电-卓越人工智能引领者 (SAIL 奖)	世界人工智能大会
2023	华为云公有云总裁奖	华为技术有限公司
2022	华为中央研究院总裁奖	华为技术有限公司
2021	Academic Travel Award	佛罗里达州立大学
2020	ICDM Student Travel Award	IEEE ICDM
2020	AAAI Student Travel Award	AAAI DLGMA
2019	NSF Student Travel Award	IEEE TPS
2019	NSF Student Travel Award	IEEE ICDE
2013	艺术体育特长人民奖学金	南京大学
2013	人民奖学金	南京大学
2012	优秀共青团员	南京大学

教授课程

2026 春	90112101-高级程序设计. 智能科学与技术学院, 南京大学
2020 春	CIS 5930-0003: Deep Learning and Applications. 计算机学院, 佛罗里达州立大学 Co-instructor: Jiawei Zhang, Ph.D
2019 春	CIS 5930-0002: Advanced Data Mining. 计算机学院, 佛罗里达州立大学 Co-instructor: Jiawei Zhang, Ph.D
2018 春	CIS 5930-0004: Social Network Mining. 计算机学院, 佛罗里达州立大学 Co-instructor: Jiawei Zhang, Ph.D

在研项目

2026-2027	AI 驱动集成电路设计导论. ◦ PI, 南京大学 AI+X 研究生课程专项
2025-2026	面向光子晶体面发射激光器 (PCSEL) 的 AI 辅助设计研究. ◦ Co-PI, 南京大学 Chip-X 未来智能芯片交叉研究中心 AI4S 专项项目
2025-2026	大模型长文本能力在个性化交互场景下的训练与应用技术研究. ◦ Co-PI, 南京大学-中国移动联合研究院研发项目

学术论文

发表论文	会议论文 : 27, 期刊论文 : 5
在投论文	提交会议 : 3, 提交期刊 : 2

会议论文

*: 共同一作 ◇: 通讯作者

- [C1] Junyu Chen, Junzhuo Li, Zhen Peng, Wenjie Wang, **Yuxiang Ren**, Long Shi, Xuming Hu. LoTA-QAF: Lossless Ternary Adaptation for Quantization-Aware Fine-Tuning. In: Proceedings of The Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS'25)

(CCF-A)

- [C2] Yuhan Li, Yuyao Wang, Jianheng Tang, Heng Chang, **Yuxiang Ren**, Jia Li. Advancing Graph Foundation Models: A Data-Centric Perspective. In: Proceedings of the ACM SIGKDD Conference on Knowledge Discovery and Data Mining 2025 (SIGKDD '25)(CCF-A)
- [C3] Wenhao Liu, Zhenyi Lu, Xinyu Hu, Jerry Zhang, Dailin Li, Jiacheng Cen, Huilin Cao, Haiteng Wang, Yuhan Li, Xie Kun, Dandan Li, Pei Zhang, Chengbo Zhang, **Yuxiang Ren**[◇], Yan Ma, Xiaohong Huang. STORM-BORN: A Challenging Mathematical Derivations Dataset Curated via a Human-in-the-Loop Multi-Agent Framework. In: Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (ACL '25 Findings). (CCF-A)
- [C4] Shengyin Sun*, **Yuxiang Ren**[◇], Jiehao Chen, and Chen Ma. Large Language Models as Topological Structure Enhancers for Text-Attributed Graphs. In: Proceedings of The 30th International Conference on Database Systems for Advanced Applications (DASFAA '25 Oral). (CCF-B)
- [C5] Yuhan Li, Xinni Zhang, Linhao Luo, Heng Chang, **Yuxiang Ren**, Irwin King, Jia Li. G-Refer: Graph Retrieval-Augmented Large Language Model for Explainable Recommendation. In: Proceedings of The Web Conference 2025 (WWW '25 Oral).(CCF-A)
- [C6] Jianxiang Yu, **Yuxiang Ren**[◇], Chenghua Gong, Jiaqi Tan, Xiang Li and Xuecang Zhang. Using LLMs for Node Generation in Few-shot Learning on Text-Attributed Graphs. In: Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence(AAAI '25). (CCF-A)
- [C7] Shengyin Sun*, Wenhao Yu*, **Yuxiang Ren**[◇], Weitao Du, Liwei Liu, Xuecang Zhang, Ying Hu and Chen Ma. GDiffRetro: Retrosynthesis Prediction with Dual Graph Enhanced Molecular Representation and Diffusion Generation. In: Proceedings of the Thirty-Eighth AAAI Conference on Artificial Intelligence(AAAI '25 Oral). (CCF-A)
- [C8] Tianyu Zhang*, **Yuxiang Ren**[◇], Chengbin Hou, Hairong Lv, and Xuegong Zhang, Molecular Graph Representation Learning Integrating Large Language Models with Domain-specific Small Models. In: Proceedings of International Conference on Bioinformatics and Biomedicine (IEEE BIBM '24). (CCF-B)
- [C9] Jiacheng Cen, Anyi Li, Ning Lin, **Yuxiang Ren**, Wenbing Huang. Are High-Degree Representations Really Unnecessary in Equivariant Graph Neural Networks? In: Proceedings of The Thirty-eighth Annual Conference on Neural Information Processing Systems (NeurIPS'24) (CCF-A)
- [C10] Wanhao Zhang, Qianli Zhang, Enyu Yu, **Yuxiang Ren**[◇], Yeqing Meng, Mingxi Qiu and Jilong Wang. Leveraging RAG-Enhanced Large Language Model for Semi-Supervised Log Anomaly Detection. In: Proceedings of the 35th IEEE International Symposium on Software Reliability Engineering(ISSRE '24). (CCF-B)
- [C11] Haoran Sun, Maokun Li, **Yuxiang Ren**, Fan Yang, Shenheng Xu. Fast Electromagnetic Scattering Calculation for UAV Targets Based on Graph Neural Networks. In: 2024 International Applied Computational Electromagnetics Society Symposium(ACES-China 2024), Xi'an, China, August 16-19, 2024.
- [C12] Wanhao Zhang, Qianli Zhang, Enyu Yu, **Yuxiang Ren**[◇], Yeqing Meng, Mingxi Qiu and Jilong Wang. LogRAG: Semi-Supervised Log-based Anomaly Detection with Retrieval-Augmented Generation. In: Proceedings of 2024 IEEE International Conference on Web Services(IEEE ICWS'24), Shenzhen, China, July 7-13, 2024. (CCF-B)
- [C13] Tianyu Zhang*, **Yuxiang Ren**[◇], Wenzheng Feng, Weitao Du, Xuecang Zhang. CGCL: Collaborative Graph Contrastive Learning without Handcrafted Graph Data Augmentations. In:

- Proceedings of the 29th International Conference on Database Systems for Advanced Applications (*DASFAA '24*), Gifu, Japan, July 2-5, 2024. (**CCF-B**)
- [C14] Kailong Wu, Xie Yule, Jiaxin Ding, **Yuxiang Ren**, Luoyi Fu, Xinbing Wang, Chenghu Zhou. Characterizing the Influence of Topology on Graph Learning Tasks In: Proceedings of the 29th International Conference on Database Systems for Advanced Applications (*DASFAA '24*), Gifu, Japan, July 2-5, 2024. (**CCF-B**)
 - [C15] Enyu Yu, Hui Dong, **Yuxiang Ren**[◇], Minzhi Yan, Xuecang Zhang, Yi Yang, Le Yue, Zhengbin Huang. HRCA: A Heterogeneous Graph-based Adaptive Root Cause Analysis Framework In: Proceedings of 2023 IEEE 34th International Symposium on Software Reliability Engineering(*ISSRE '23 Industry Track*), Florence, Italy, October 9-12, 2023. (**CCF-B**)
 - [C16] Lin Meng, **Yuxiang Ren** and Jiawei Zhang. Decoupling Graph Neural Network with Contrastive Learning for Fraud Detection. In: Proceedings of the 28th International Conference on Database Systems for Advanced Applications (*DASFAA '23*), Tianjin, China, April 17-20, 2023. (**CCF-B**)
 - [C17] **Yuxiang Ren***, Jiyang Bai* and Jiawei Zhang. Label Contrastive Coding based Graph Neural Network for Graph Classification. In: Proceedings of the 26th International Conference on Database Systems for Advanced Applications (*DASFAA '21*), Taiwan, China, April 11-14, 2021. (**CCF-B**)
 - [C18] **Yuxiang Ren**, Hao Zhu, Jiawei Zhang, Liefeng Bo, and Peng Dai. EnsemFDet: An Ensemble Approach to Fraud Detection based on Bipartite Graph. In: Proceedings of the 2021 IEEE International Conference on Data Engineering (*ICDE '21*), Chania, Greece, April 19-22, 2021. (**CCF-A**)
 - [C19] **Yuxiang Ren** and Jiawei Zhang. Fake News Detection on News-Oriented Heterogeneous Information Networks through Hierarchical Graph Attention. In: Proceedings of the 2021 International Joint Conference on Neural Networks (*IJCNN '21*), Virtual Event, July 18-22, 2021. (**CCF-C**)
 - [C20] Jiyang Bai*, **Yuxiang Ren*** and Jiawei Zhang. Ripple Walk: A sampling based Graph Neural Network Optimization Framework. In: Proceedings of the 2021 International Joint Conference on Neural Networks (*IJCNN '21*), Virtual Event, July 18-22, 2021. (**CCF-C**)
 - [C21] Jiyang Bai, **Yuxiang Ren** and Jiawei Zhang. BGADAM: Boosting based Genetic-Evolutionary ADAM for Convolutional Neural Network Optimization. In: Proceedings of the 2021 International Joint Conference on Neural Networks (*IJCNN '21*), Virtual Event, July 18-22, 2021. (**CCF-C**)
 - [C22] **Yuxiang Ren**, Bo Wang, Jiawei Zhang and Yi Chang. Adversarial Active Learning based Heterogeneous Graph Neural Network for Fake News Detection. In: Proceedings of the 2020 IEEE International Conference on Data Mining (*ICDM '20*), Sorrento, Italy, November 17-20, 2020. (**CCF-B**)
 - [C23] **Yuxiang Ren**, Lin Meng and Jiawei Zhang. Scalable Heterogeneous Social Network Alignment through Synergistic Graph Partition. In: Proceedings of 31st ACM Conference on Hypertext and Social Media (*ACM HT '20*), Virtual Edition, July 13-15 2020.
 - [C24] **Yuxiang Ren**, Bo Liu, Peng Dai, Jiawei Zhang, Chao Huang and Liefeng Bo. HDGI: An Unsupervised Graph Neural Network for Representation Learning in Heterogeneous Graph. AAAI-20 Workshop on "Deep Learning on Graphs: Methodologies and Applications (AAAI-DLGM)", New York, USA, February 7-12, 2020.
 - [C25] Jiyang Bai, **Yuxiang Ren** and Jiawei Zhang. DEAM: Adaptive Momentum with Discriminative Weight for Stochastic Optimization. In: Proceedings of the 2020 IEEE/ACM International

Conference on Advances in Social Network Analysis and Mining (ASONAM '20), Virtual Edition, Dec 7-10 2020.

- [C26] **Yuxiang Ren**, Charu C. Aggarwal and Jiawei Zhang. Meta Diagram based Active Social Networks Alignment. In: Proceedings of the 2019 IEEE International Conference on Data Engineering (ICDE '19), Macau SAR, China, April 8-12, 2019. (CCF-A)
- [C27] Lin Meng, **Yuxiang Ren**, Jiawei Zhang, Fanghua Ye and Philip S. Yu. Deep Heterogeneous Social Network Alignment. In: Proceedings of the 2019 IEEE International Conference on Cognitive Machine Intelligence (IEEE CogMI '19), Los Angeles, USA, December 12-14, 2019. (Invited Paper)

期刊论文

*: 共同一作 ◇: 通讯作者

- [J1] Huilin Cao, Haoran Sun, Jingwei Liang, Tao Shan, **Yuxiang Ren**◇, Maokun Li. Physics-Data Hybrid Driven Edge-Featured Graph Attention Networks for Surface Currents Learning in EM Scattering of 3D PEC Targets. IEEE Transactions on Antennas and Propagation, 2025. (SCI Q1)
- [J2] Kaiwei Zhang, Yange Lin , Guangcheng Wu, **Yuxiang Ren**◇, Xuechang Zhang, Bo Wang, Xiaoyu Zhang, Weitao Du. Sculpting Molecules in Text-3D Space: A Flexible Substructure Aware Framework for Text-Oriented Molecular Optimization. BMC BIOINFORMATICS, 2025. (SCI Q1)
- [J3] **Yuxiang Ren**◇, Charu C. Aggarwal and Jiawei Zhang. *ActiveIter*: Meta Diagram based Active Learning in Social Networks Alignment. IEEE Transactions on Knowledge and Data Engineering (TKDE), 2019. (SCI Q1, CCF-A)
- [J4] Jiyang Bai*, **Yuxiang Ren***◇ and Jiawei Zhang. Adaptive Momentum with Discriminative Weight for Neural Network Stochastic Optimization. International Journal of Intelligent Systems, 2022. (SCI Q1)
- [J5] Jiyang Bai*, **Yuxiang Ren***◇ and Jiawei Zhang. Measuring and Sampling: A Metric-guided Subgraph Training Framework for Graph Neural Network. International Journal of Intelligent Systems, 2022. (SCI Q1)

授权专利

- 2025 CN120636507A: 杜伟韬, 任宇翔, 陈玖玖. 一种分子表征信息生成模型的训练方法及相关装置
- 2025 CN119441586B: 石川, 任宇翔, 刘佳玮. 一种推荐方法及装置.
- 2024 CN118133031A: 石川, 任宇翔, 闫博. 异常事件检测方法、异常事件检测模型构建方法以及装置.
- 2023 US20230112441A1: LuAn Tang, Yuncong Chen, Wei Cheng, Zhengzhang Chen, Haifeng Chen, Yuji Kobayashi, **Yuxiang Ren**. Data fusion and analysis engine for vehicle sensors
- 2022 US20220111836A1: LuAn Tang, Wei Cheng, Haifeng Chen, Zhengzhang Chen, **Yuxiang Ren**. Modular network based knowledge sharing for multiple entities
- 2020 CN111652837A: 张航, 任宇翔, 赵佳琦, 章建全. 基于 AI 的甲状腺结节左右叶定位与超声报告纠错方法.

指导学生

- 2023–2025 曹慧琳.
- 博士研究生 @ 上海交通大学
 - 研究课题: AI for Science, GNN4PDE
 - 合作论文: [J1]??
- 2024–2025 李宇涵.
- 博士研究生 @ 香港科技大学
 - 研究课题: Graph RAG, Graph for LLM
 - 合作论文: [C5][C2]
- 2024–2025 刘文豪.
- 硕士研究生 @ 北京邮电大学
 - 研究课题: LLM4Math, LLM Benchmark
 - 合作论文: [C3]
- 2023–2024 虞健翔.
- 博士研究生 @ 华东师范大学
 - 研究课题: Graph Representation Learning
 - 合作论文: [C6]
- 2023–2024 于文昊.
- 硕士研究生 @ 香港中文大学
 - 研究课题: AI for Science
 - 合作论文: [C7]
- 2023–2024 孙圣印.
- 博士研究生 @ 香港城市大学
 - 研究课题: AI for Science
 - 合作论文: [C4][C7]
- 2023–2024 张天宇.
- 硕士研究生 @ 清华大学
 - 研究课题: Graph Representation Learning, AI for Science
 - 合作论文: [C8][C13]
- 2023–2024 张万豪.
- 硕士研究生 @ 清华大学
 - 研究课题: Artificial Intelligence for IT Operations
 - 合作论文: [C10][C12]
- 2023–2024 吴凯龙.
- 硕士研究生 @ 上海交通大学
 - 研究课题: Graph Neural Network, Topology Optimization
 - 合作论文: [C14]
- 2018–2022 白霁阳.
- 研究员 @ 华为泊松实验室
 - 研究课题: Graph Neural Network, Stochastic Optimization
 - 合作论文: [C17],[C20],[C21],[C25],[J4],[J5]

- 2018–2022 孟琳.
- 研究员 @ VISA Research Center
 - 研究课题: Social Network Alignment
 - 合作论文: [C16], [C23], [C27]
- 2018–2019 王博.
- 博士研究生 @ 吉林大学
 - 研究课题: Fake News Detection
 - 合作论文: [C22]

学术服务

国家标准制定

- 全国信息技术标准化技术委员会国家标准《信息技术神经网络表示与模型压缩第 3 部分：图神经网络》提案起草 (排名第四)

期刊编委

- 《AI for Science》期刊青年编委, 2026-2028

学术会议审稿

- Learning on Graphs Conference, *LoG* 2024
- IEEE International Conference on Bioinformatics and Biomedicine, *IEEE BIBM* 2024
- International Conference on Machine Learning, *ICML* 2022
- SIGKDD Conference of Knowledge Discovery and Data Mining, *KDD* 2022, 2023, 2024
- The Web Conference, *WWW* 2023, 2024
- IEEE World Congress on Computational Intelligence, *IEEE WCCI* 2022
- The International Conference on Learning Representations, *ICLR* 2022, 2023
- The Conference on Neural Information Processing Systems, *NeurIPS* 2020, 2021
- Thirty-Fifth AAAI Conference on Artificial Intelligence *AAAI* 2021, 2023
- The European Conference on Machine Learning and Principles and Practice of Knowledge Discovery in Databases, Research Track, *ECML-PKDD* 2020

期刊审稿

- IEEE Transactions on Pattern Analysis and Machine Intelligence 2025
- IEEE Transactions on Industrial Informatics 2025
- IEEE Transactions on Cybernetics 2025
- Journal of Chemical Information and Modeling 2025
- ACM Transactions on Knowledge Discovery from Data, *TKDD* 2025
- IEEE Transactions on Knowledge and Data Engineering, *TKDE* 2020, 2021, 2025
- IEEE Transactions on Neural Networks and Learning Systems, *TNNLS* 2021, 2022
- IEEE Transactions on Big Data *TBD* 2020, 2021, 2025

- Cybernetics and Systems 2022
- Computer Networks 2024
- IEEE Sensors Journal 2025